AI Workspace — M4 Starter

This starter contains only the shared project baseline:

- the M1/M2 backend headers and the M3 GUI headers;
- `main_window.hpp` with the two public functions M4 requires (spec Section 5);
- a CMake configuration with the required targets `ai_workspace_m4` and `student_tests`;
- the public sample data;
- empty `DESIGN_NOTE.md` and `DESIGN.md` files.

Nothing for the new M4 work is provided. Designing it is the assignment.

Getting started

1. Copy your completed M3 `src/` files into `src/`. If your M3 `main_window.hpp` differs from the starter's, keep your own and add the two M4 declarations from the starter's version (spec Section 5).

2. Inside the course Docker container, build it once, unchanged, to confirm your M3 baseline works:

   ```
   cd /mnt && mkdir build && cd build
   cmake .. && make && ctest --output-on-failure
   ```

3. Read the specification, then write `DESIGN_NOTE.md` (due Wed Oct 28) before most of your code.

Over the container's X display, windows redraw slowly. Slow is normal; a window that freezes during a build is not.

Commit often. `HISTORY.txt` (spec Section 8) comes from your git log.